

Employee Attrition Prediction

Summary Report for HR Leadership
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What This Report Is About

This project used employee data from IBM's HR records (1,470 employees) to build a machine learning system that can predict which employees are at risk of leaving the company before they actually resign. The goal is to give the HR team enough advance warning to step in with the right support — whether that's a salary review, a workload adjustment, or a one-on-one conversation.

Key Numbers at a Glance

Metric	Value
Total employees in dataset	1,470
Employees who left	237 (16.1%)
Employees who stayed	1,233 (83.9%)
Department with highest exits	Sales (20.6%)
Job role with highest exits	Sales Representative (39.8%)
Avg. income of those who left	Rs. 4,787 lower than those who stayed
Most exits happen at tenure	First 1-3 years

What the Model Found: Top 3 Exit Triggers

After analysing all available data, the system identified three factors that most strongly predict whether an employee will leave:

- Overtime:** Employees who regularly work overtime are the highest-risk group. This was the single strongest signal in the entire dataset — stronger even than salary.
- Monthly Income:** Employees earning below the department average are significantly more likely to leave. Lower pay combined with overtime is an especially dangerous combination.
- Job Level and Age:** Younger employees in junior roles (Job Level 1-2) leave far more frequently. Most exits happen within the first 3 years of joining.

Where HR Should Focus First

The Sales department — specifically Sales Representatives — needs immediate attention. Nearly 4 out of every 10 Sales Representatives left in this dataset. This is almost double the

company-wide rate. The combination of high targets, variable pay, and frequent overtime makes this group the most vulnerable.

Is It Just About Salary?

No. While pay matters, overtime emerged as a stronger predictor than salary alone. Employees who are well-paid but consistently overworked still leave at high rates. Work-life balance ratings and the number of years since the last promotion also played significant roles. This means that salary increases alone will not solve the attrition problem — the underlying workload and career growth issues must also be addressed.

Two Immediate Recommendations

Recommendation 1 — Overtime Alert System: Any employee who has been working overtime for more than 3 consecutive months, particularly in Sales or junior roles, should be automatically flagged for an HR check-in. This conversation should explore workload, satisfaction, and near-term career plans — not wait for a resignation letter.

Recommendation 2 — Early Tenure Programme: Since most exits happen in the first 1-3 years, introduce a structured engagement programme for all new hires. This should include formal check-ins at 6, 12, and 18 months, a dedicated mentor, and a clear career progression discussion. Catching dissatisfaction early is far cheaper than replacing an employee.

Important Note for HR Leadership

This model is a risk-flagging tool, not a final verdict. It identifies employees who share characteristics with people who have left in the past. HR should use these predictions to start conversations — not to make decisions about individuals. The model should also be updated every 6-12 months to stay accurate as the workforce changes.